

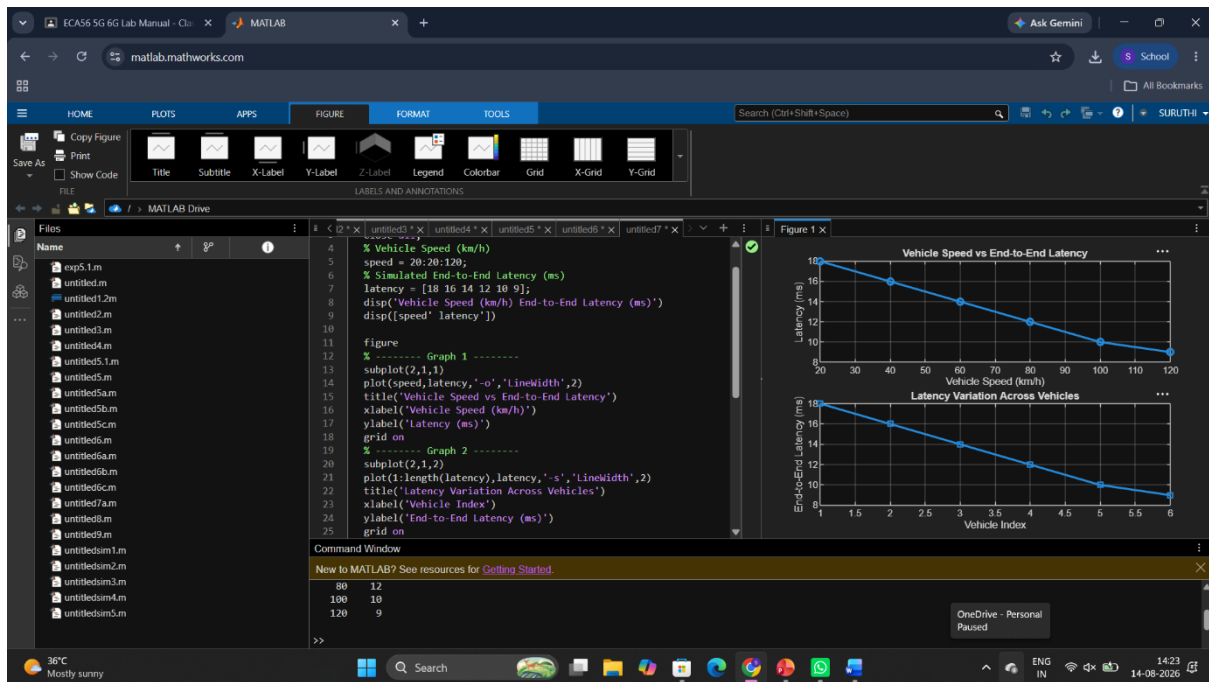
Ex.No. 13: V2X Latency Comparison

Objective1:

Program:

```
clc;
clear;
close all;
% Vehicle Speed (km/h)
speed = 20:20:120;
% Simulated End-to-End Latency (ms)
latency = [18 16 14 12 10 9];
disp('Vehicle Speed (km/h) End-to-End Latency (ms)')
disp([speed' latency'])
figure
...xlabel('Vehicle Speed (km/h)')
ylabel('Latency (ms)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(1:length(latency),latency,'-s','LineWidth',2)
title('Latency Variation Across Vehicles')
xlabel('Vehicle Index')
ylabel('End-to-End Latency (ms)')
grid on
```

output:



Objective2:

Program:

clc;

clear;

close all;

% Vehicle Speed (km/h)

speed = 20:20:120;

% Simulated Packet Delivery Ratio (%)

PDR = [99 98 97 95 93 90];

disp('Vehicle Speed (km/h) Packet Delivery Ratio (%)')

disp([speed' PDR'])

figure

...xlabel('Vehicle Speed (km/h)')

ylabel('Packet Delivery Ratio (%)')

grid on

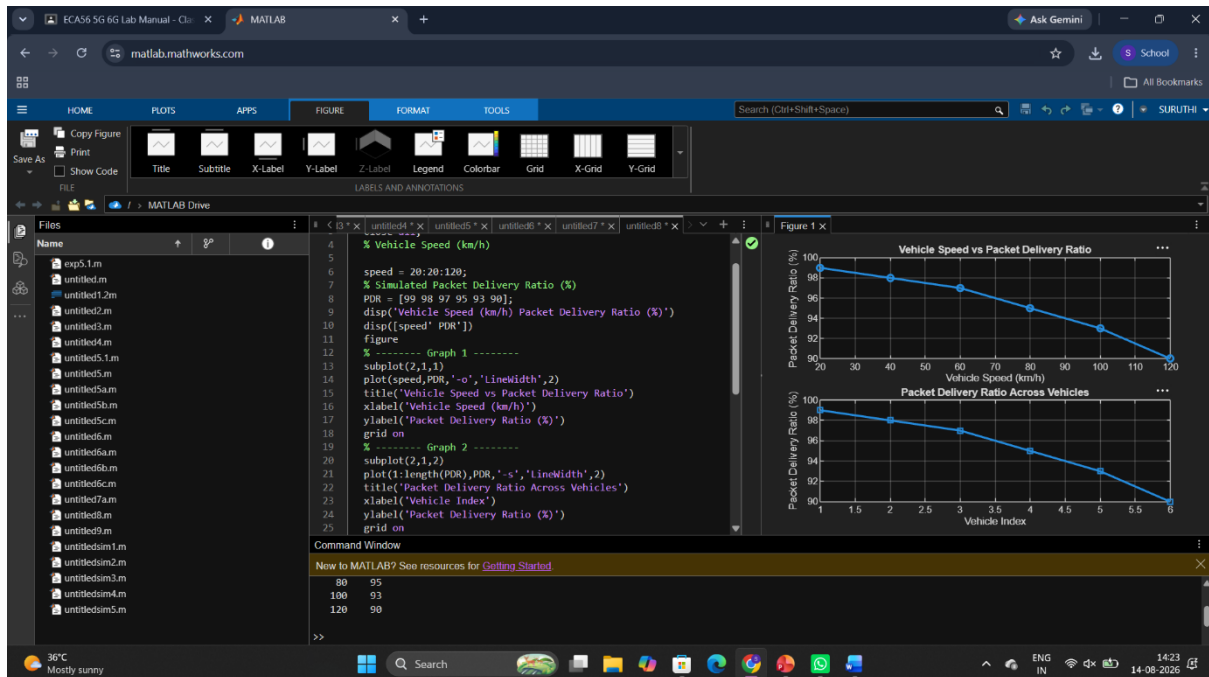
% ----- Graph 2 -----

subplot(2,1,2)

```

plot(1:length(PDR),PDR,'-s','LineWidth',2)
title('Packet Delivery Ratio Across Vehicles')
xlabel('Vehicle Index')
ylabel('Packet Delivery Ratio (%)')
grid on
output:

```



Objective3:

Program:

```

clc;
clear;
close all;
% Vehicle IDs
vehicle = 1:6;
% Vehicle Speed (km/h)
speed = [30 45 60 75 90 105];
% Communication Range (m)
range = [280 260 240 220 200 180];

```

```
disp('Vehicle Speed(km/h) Communication Range(m)')
```

```
...xlabel('Vehicle Index')
```

```
ylabel('Vehicle Speed (km/h)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

```
plot(speed,range,'-s','LineWidth',2)
```

```
title('Vehicle Speed vs Communication Range')
```

```
xlabel('Vehicle Speed (km/h)')
```

```
ylabel('Communication Range (m)')
```

```
grid on
```

output:

